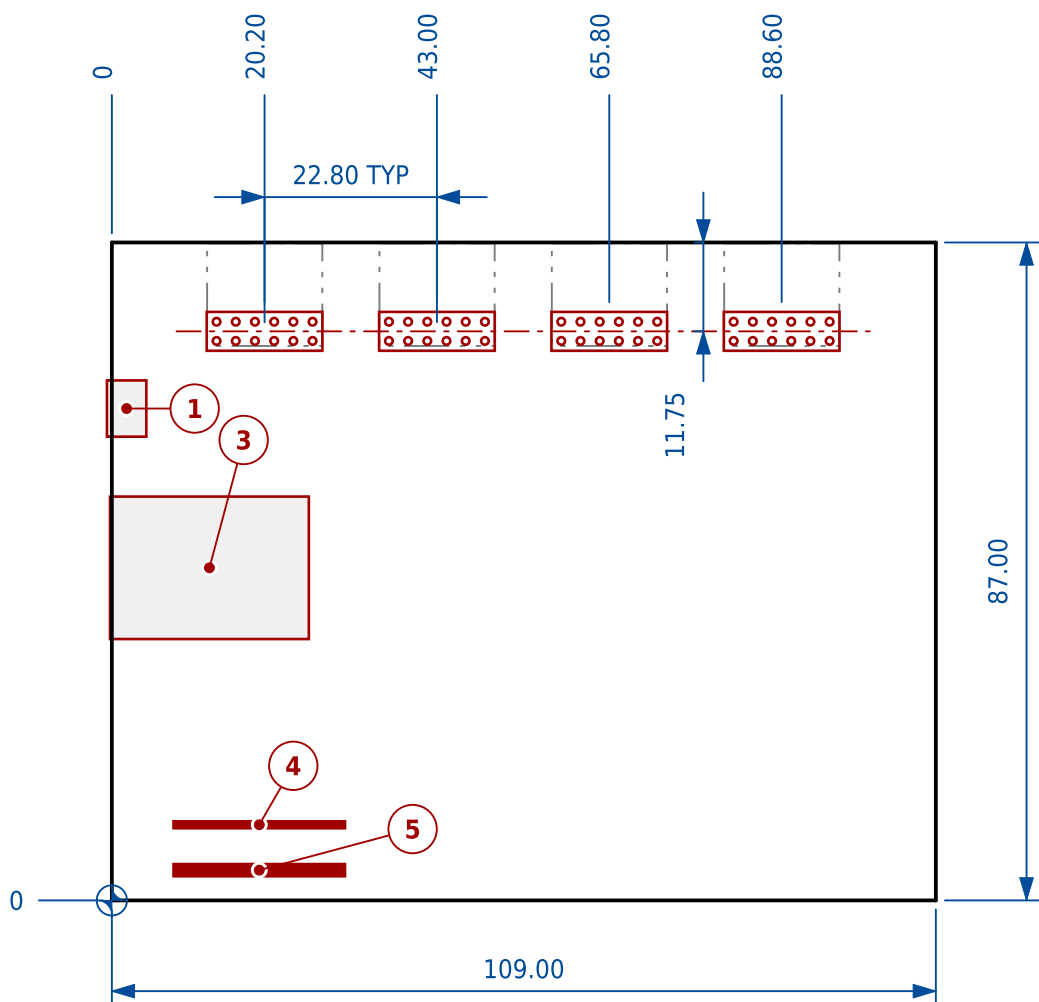




PMOD HOST HEADERS

PORT	EDGE	CX mm	CY mm	PIN1 X mm	PIN1 Y mm
JA	top	20.200	75.250	13.850	74.000
JB	top	43.000	75.250	36.650	74.000
JC	top	65.800	75.250	59.450	74.000
JD	top	88.600	75.250	82.250	74.000

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	micro-USB J10, programming and console	-0.65 to 4.56	61.32 to 68.76
2	Second USB port - not on this board	-	-
3	Ethernet RJ45 J9	-0.24 to 26.06	34.56 to 53.40
4	LEDs LD4-LD7, 4 on 7.00 mm pitch	8.16 to 30.84	9.54 to 10.46
5	Tri-colour LEDs LD0-LD3, 4 on 7.00 mm pitch	8.16 to 30.84	3.19 to 4.80
6	Expansion connector, first - not on this board	-	-
7	Expansion connector, second - not on this board	-	-
8	Expansion connector, third - not on this board	-	-
9	Third USB port - not on this board	-	-
10	Fourth USB port - not on this board	-	-
11	Status LEDs, debug controller - not on this board	-	-

LEGEND

- Board outline
- Component body, scheduled
- Pmod connector body
- Dimension, extension and leader

NOTES

- Viewed from the component side.
- No mounting holes. The drawing shows four 10.0 mm rubber feet centred 5.0 mm from each corner; a plate has to carry the board on those or clamp its edges.
- The Pmod pin rows are 2.50 mm apart on the drawing, not 2.54, and the hosts are on a 22.80 mm pitch where the PDF's dimension reads 0.90 in; both within the general tolerance.
- JA is nearest the USB and Ethernet corner, per the reference manual's callout figure.
- The 1.6 x 1.6 mm LED row nearest the edge is taken as the tri-colour LD0-LD3 and the 0603 row behind it as LD4-LD7; no source names them.
- Bodies read from the PDF plot are good to about +/-0.3 mm; the DXF figures carry the general tolerance.
- Assembled envelope, connector overhang included: 109.65 x 87.00 mm.

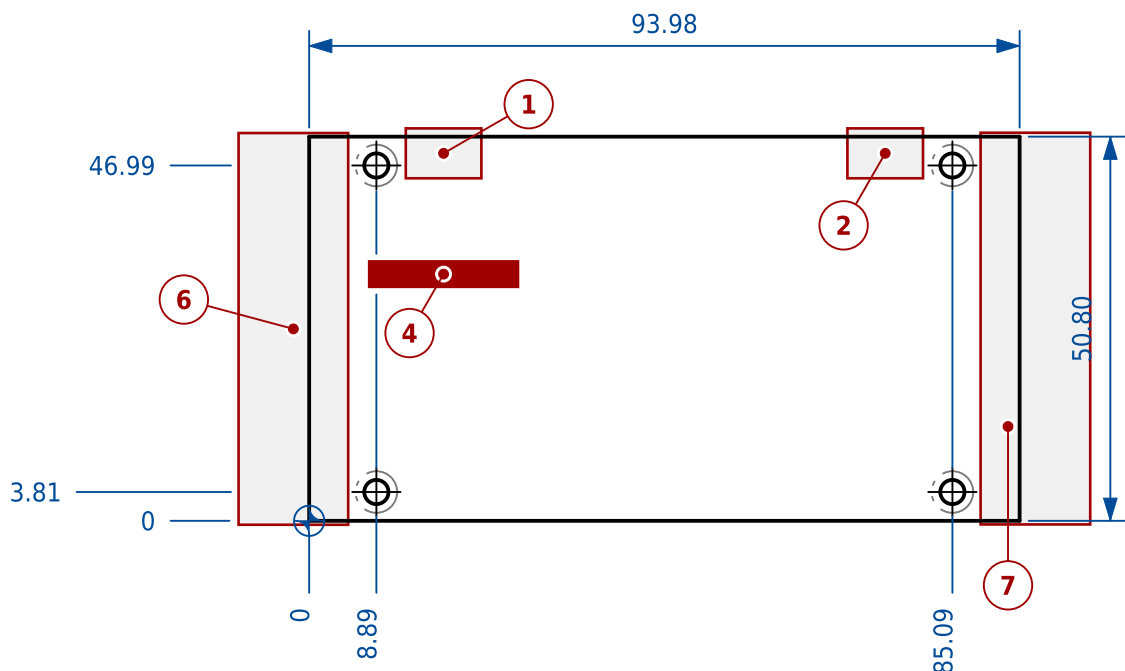
SOURCES

- Mechanical drawing: https://digilent.com/reference/_media/reference/programmable-logic/arti-a7/arti_a7.zip - Digilent, Arty_A7_DXF.DXF and Mechanical_Arty_A7.pdf, dated 2020-09-02. Outline, pin fields, slots and pegs from the DXF; component bodies from the PDF, whose plot scale is recovered from the outline (axes agree to 0.20 %).
- 3D model: https://digilent.com/reference/_media/reference/programmable-logic/arti/arti_rev_cad.zip - Digilent, Arty Rev C: used only to identify which of the PDF's outlines are the eight user LEDs.
- Reference manual: <https://digilent.com/reference/programmable-logic/arti-a7/reference-manual> - Pmod JA, JB, JC, JD; four tri-colour and four single LEDs.
- Mounting: <https://forum.digilent.com/topic/4523-410-319-mounting/> - Digilent staff, 2017-07-19: "There are no thru holes or mounting provisions"; the board stands on rubber feet.

DIGILENT ARTY A7

A7-35T and A7-100T

DRAWN generated		VERSION v0.0-270-gc2d9e2f	UNITS mm	SCALE 1:1	
SIZE A3	SHEET 1 OF 1	DRAWING NO FPGA-ARTY-A7			REV A
MATERIAL PCB, thickness not stated		GENERAL TOLERANCE edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08			



HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
H1	8.89	3.81	3.20	5.50
H2	85.09	3.81	3.20	5.50
H3	85.09	46.99	3.20	5.50
H4	8.89	46.99	3.20	5.50

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	micro-USB US1, FTDI programming and console	12.78 to 22.78	45.30 to 51.90
2	micro-USB US2, USB host / OTG	71.20 to 81.20	45.30 to 51.90
3	Ethernet RJ45 - not on this board	-	-
4	LEDs D0-D7, 8 on 2.54 mm pitch	7.94 to 27.62	30.94 to 34.30
5	User LEDs, tri-colour - not on this board	-	-
6	GPIO header J1, 2x20	-9.34 to 5.16	-0.52 to 51.28
7	GPIO header J2, 2x20	88.82 to 103.32	-0.48 to 51.32
8	Expansion connector, third - not on this board	-	-
9	Third USB port - not on this board	-	-
10	Fourth USB port - not on this board	-	-
11	Status LEDs, debug controller - not on this board	-	-

LEGEND

- Board outline
- Component body, scheduled
- Hole keep-out
- Dimension, extension and leader

NOTES

- Viewed from the component side.
- Hole IDs are the KiCad reference designators.
- No Pmod host. J1 and J2 are right-angle 2x20 sockets on 2.54 mm whose pinout is Pmod compatible, per the maker; pin 1 of each is given in the schedule note.
- No Ethernet jack.
- Assembled envelope, connector overhang included: 112.66 x 52.42 mm.

SOURCES

- KiCad board file: https://github.com/emard/ulx3s/ulx3s.kicad_pcb @ 300a361b (tag v3.0.3) - Radiona / emard
- KiCad board file: https://github.com/emard/ulx3s/ulx3s.kicad_pcb @ d518b2ce (tag v3.0.7) - Radiona / emard

SOURCES (continued)

- KiCad board file: https://github.com/emard/ulx3s/ulx3s.kicad_pcb @ 9908a1f8 (tag v3.0.8) - Radiona / emard
- KiCad board file: https://github.com/emard/ulx3s/ulx3s.kicad_pcb @ cc1817d2 (tag v3.1.7) - Radiona / emard
- Board versions: <https://github.com/emard/ulx3s/blob/master/doc/MANUAL.md> - section "Board Versions": v3.0.3, v3.0.7, v3.0.8 and v3.1.7 are the rows marked "for sale"; the four board files are read and agree on everything drawn here.
- Board size: <https://github.com/emard/ulx3s/blob/master/README.md> - quoted: "This is a small (94x51 mm) standalone FPGA board"; the board file gives 93.98 x 50.80, which is 37 x 20 tenths of an inch.

ULX3S

v3.0.3, v3.0.7, v3.0.8 and v3.1.7

DRAWN generated		VERSION v0.0-400-g80d77dd		UNITS mm		SCALE 1:1	
SIZE A3	SHEET 1 OF 1		DRAWING NO FPGA-ULX3S				REV A
MATERIAL PCB, 1.6 nominal			GENERAL TOLERANCE edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08				

A

B

C

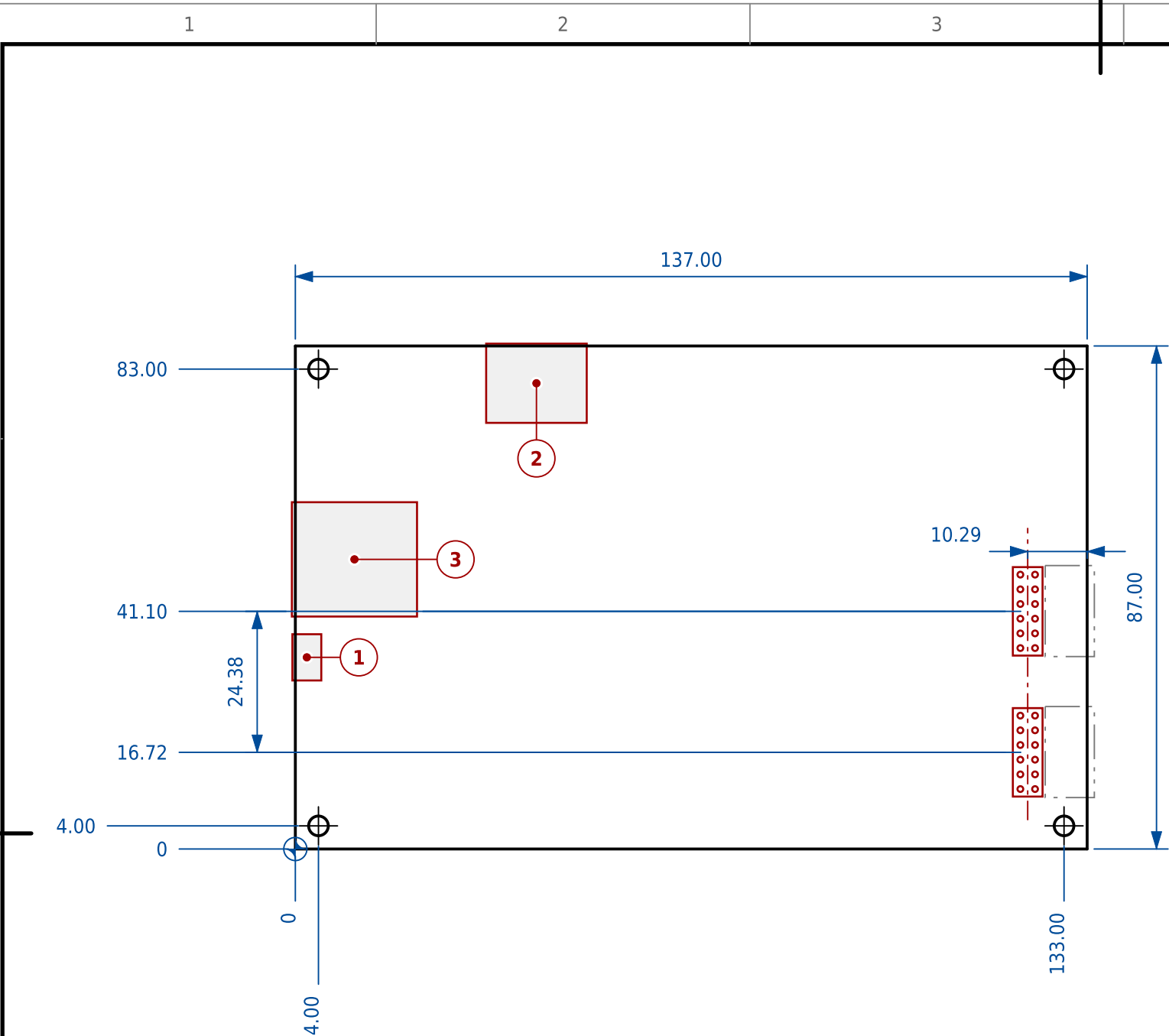
D

A

B

C

D



HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
MT1	4.00	4.00	3.40	not given
MT2	133.00	4.00	3.40	not given
MT3	4.00	83.00	3.40	not given
MT4	133.00	83.00	3.40	not given

PMOD HOST HEADERS

PORT	REF	EDGE	CX mm	CY mm	PIN1 X mm	PIN1 Y mm
Pmod A	JA	right	126.713	41.101	125.443	47.451
Pmod B	JB	right	126.713	16.717	125.443	23.067

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	micro-USB, programming and console	-0.51 to 4.49	29.15 to 37.15
2	USB type A host port	33.02 to 50.42	73.70 to 87.40
3	Ethernet RJ45	-0.60 to 21.05	40.20 to 59.98
4	User LEDs, single colour - not on this board	-	-
5	User LEDs, tri-colour - not on this board	-	-
6	Expansion connector, first - not on this board	-	-
7	Expansion connector, second - not on this board	-	-
8	Expansion connector, third - not on this board	-	-
9	Third USB port - not on this board	-	-
10	Fourth USB port - not on this board	-	-
11	Status LEDs, debug controller - not on this board	-	-

LEGEND

- Board outline
- Component body, scheduled
- Pmod connector body
- Dimension, extension and leader

NOTES

- Viewed from the component side.
- Every dimension is read from the STEP model; TUL publish no drawing. Hole IDs are assigned by this drawing.
- The user LEDs LD0-LD3 and the two tri-colour LEDs are not in the model, so schedule rows 4 and 5 are empty; the board has them, above the four push buttons near the lower edge.
- TUL's product sheet gives the board as 87 x 140 mm; the model's laminate is 137.0 wide and the Pmod hosts reach 1.2 beyond it.
- Assembled envelope, connector overhang included: 137.60 x 87.40 mm.

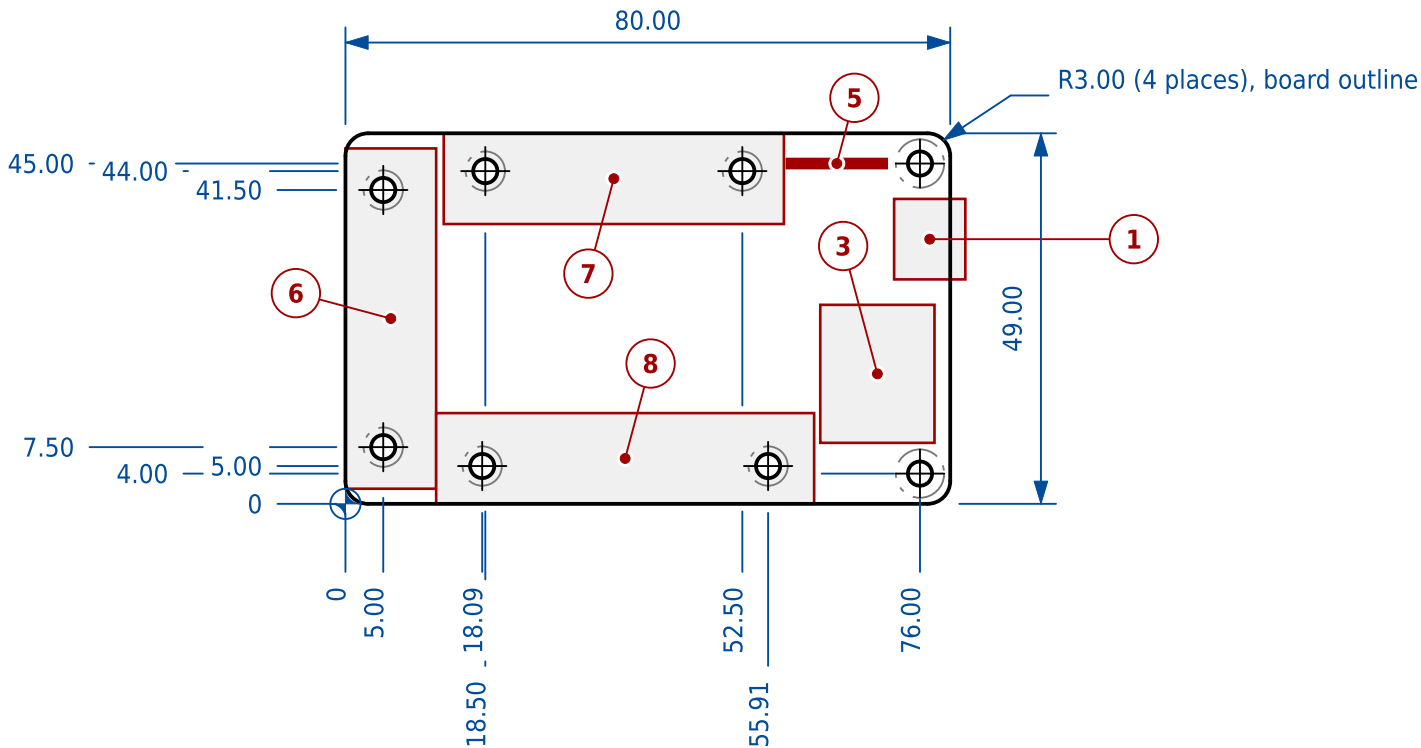
SOURCES

- 3D model:
https://discuss.pynq.io/t/pynq-z2-pcb-3d-step-files/2645-PYNQ_Z2_20220218.STEP, TUL's SolidWorks export of 2022-02-18 posted by Xilinx; the earlier model on the same thread has the Pmod spacing wrong and is not used.
- User manual:
https://dpoauwqwqsy2x.cloudfront.net/Download/PYNQ_Z2_User_Manual_v1.1.pdf - v1.1: Pmod A is the upper host (section 16); 4 LEDs and 2 tri-colour LEDs (section 14).

TUL PYNQ-Z2

137 x 87 mm

DRAWN generated		VERSION v0.0-270-gc2d9e2f		UNITS mm		SCALE 1:1	
SIZE A3	SHEET 1 OF 1		DRAWING NO FPGA-PYNQ-Z2				REV A
MATERIAL PCB, 1.6 nominal			GENERAL TOLERANCE edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08				



HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
H1	76.00	45.00	3.20	6.40
H2	76.00	4.00	3.20	6.40
SA1	5.00	7.50	3.20	5.20
SA2	5.00	41.50	3.20	5.20
SB1	18.50	44.00	3.20	5.20
SB2	52.50	44.00	3.20	5.20
SC1	18.09	5.00	3.20	5.20
SC2	55.91	5.00	3.20	5.20

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	USB-C J1, programming and console	72.58 to 82.00	29.68 to 40.32
2	Second USB port - not on this board	-	-
3	Ethernet RJ45 CON1	62.81 to 77.92	8.06 to 26.32
4	User LEDs, single colour - not on this board	-	-
5	RGB LEDs D1-D7, 7 on 2.00 mm pitch	58.40 to 71.60	44.40 to 45.60
6	SYZYGY port A, J3	0.00 to 12.00	2.00 to 47.00
7	SYZYGY port B, J4	13.00 to 58.00	37.00 to 49.00
8	SYZYGY port C, J5	12.00 to 62.00	0.00 to 12.00
9	Third USB port - not on this board	-	-
10	Fourth USB port - not on this board	-	-
11	Status LEDs, debug controller - not on this board	-	-

LEGEND

- Board outline
- Component body, scheduled
- Hole keep-out
- Dimension, extension and leader

NOTES

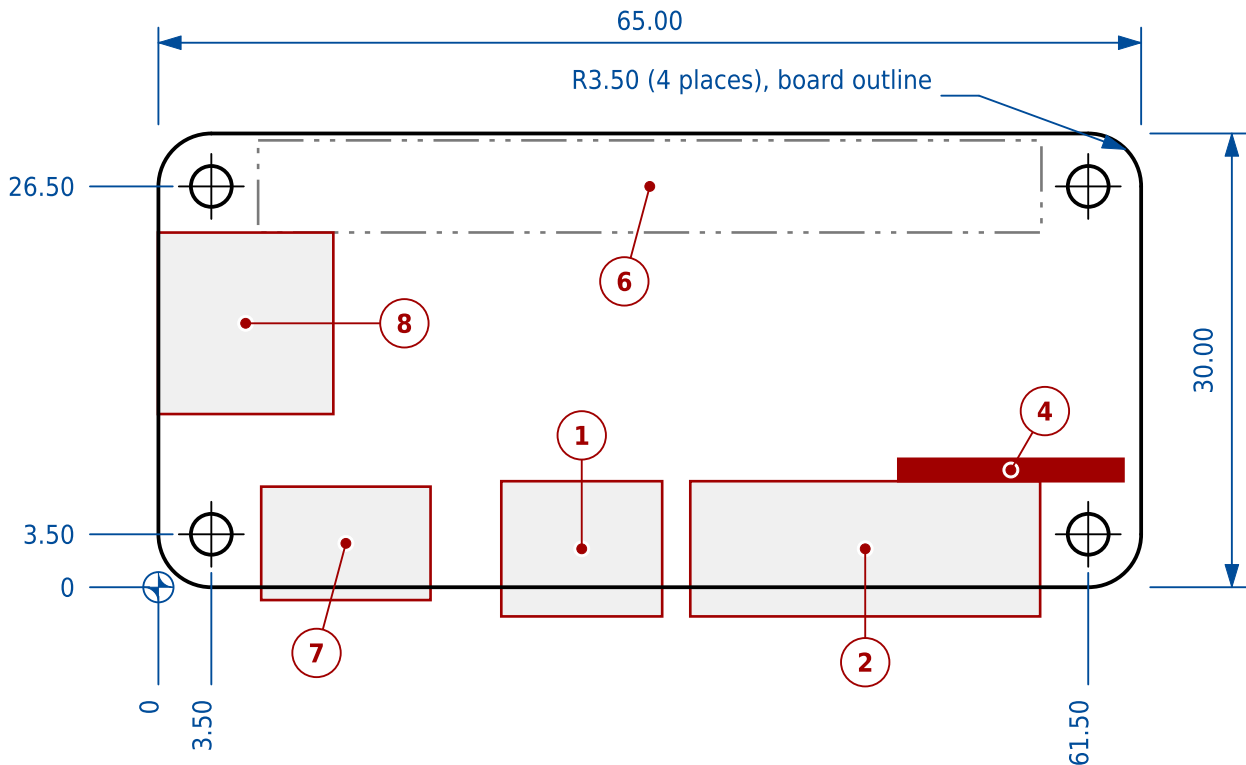
- Viewed from the component side.
- Hole IDs H1 and H2 are the KiCad reference designators; SA1 to SC2 are the SYZYGY standoff holes of ports A, B and C, plated, and the maker's acrylic plate bolts through all eight.
- No Pmod host: expansion is three SYZYGY ports.
- Assembled envelope, connector overhang included: 82.00 x 49.00 mm.

SOURCES

- KiCad board file:
https://github.com/butterstick-fpga/butterstick-hardware/blob/main/ButterStick_r1.0/ButterStick.kicad_pcb @ fac6a834 - title block: ButterStick rev r1.0a, dated 2021-01-10; Greg Davill / GsD
- Board size:
<https://github.com/butterstick-fpga/butterstick-hardware/blob/main/README.md> - quoted: "Board dimensions: 80mm x 49mm"
- Thickness:
https://github.com/butterstick-fpga/butterstick-hardware/blob/main/hardware/ButterStick_r1.0/Production/ButterStick-r1.0a-fab-notes.txt - quoted: "Finished board thickness is 1.6 mm."

BUTTERSTICK

r1.0			
DRAWN		VERSION	SCALE
generated		v0.0-400-g80d77dd	1:1
SIZE	SHEET	DRAWING NO	REV
A3	1 OF 1	FPGA-BUTTERSTICK	A
MATERIAL		GENERAL TOLERANCE	
PCB, 1.6 nominal		edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08	



HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
H3	3.50	3.50	2.70	not given
H4	61.50	3.50	2.70	not given
H1	3.50	26.50	2.70	not given
H2	61.50	26.50	2.70	not given

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	USB-C J3, JTAG and console	22.68 to 33.32	-1.93 to 7.01
2	USB-C J5 and J4, to the FPGA, 2 on 12.50 mm pitch	35.18 to 58.32	-1.93 to 7.01
3	Ethernet RJ45 - not on this board	-	-
4	User LEDs LED0-LED4, 5 on 2.98 mm pitch	48.94 to 63.83	7.01 to 8.49
5	User LEDs, tri-colour - not on this board	-	-
6	GPIO header J1, 2x20, not fitted	6.59 to 58.41	23.45 to 29.55
7	GPD video connector J2	6.80 to 18.00	-0.85 to 6.65
8	microSD card socket J6	-0.03 to 11.57	11.45 to 23.45
9	Third USB port - not on this board	-	-
10	Fourth USB port - not on this board	-	-
11	Status LEDs, debug controller - not on this board	-	-

LEGEND

- Board outline
- Component body, scheduled
- Position not fitted on this revision
- Dimension, extension and leader

NOTES

- Viewed from the component side.
- Hole IDs are the KiCad reference designators.
- The mounting holes are the Raspberry Pi Zero pattern: 58.00 x 23.00 mm, 3.50 mm in from each edge of a 65 x 30 board. They are drilled 2.70, the bottom of the Pi Zero drawing's 2.75 +/-0.05 band; the corners are R3.50 where the Pi Zero's are R3.00.
- No Pmod host and no Ethernet jack.
- The unfitted GPIO position is a Raspberry Pi 40-pin header: 1.00 mm holes on a 2.54 mm grid, with the Pi's 5 V, 3V3 and first GPIO pins in the Pi's places and pin 1 at (8.38, 25.23), in the row farther from the board edge.
- The two user buttons are on the underside, centred at (46.40, 15.45) and (46.40, 18.85); a plate under the board has to clear them.
- The board file has been re-annotated since the mass-production release. Nothing drawn here moved, but at d67bb758 the programming port is J5 and the user LEDs run D1 to D5 left to right; the designators here are the ones the sold boards were fabbed with.
- Assembled envelope, connector overhang included: 65.03 x 31.93 mm.

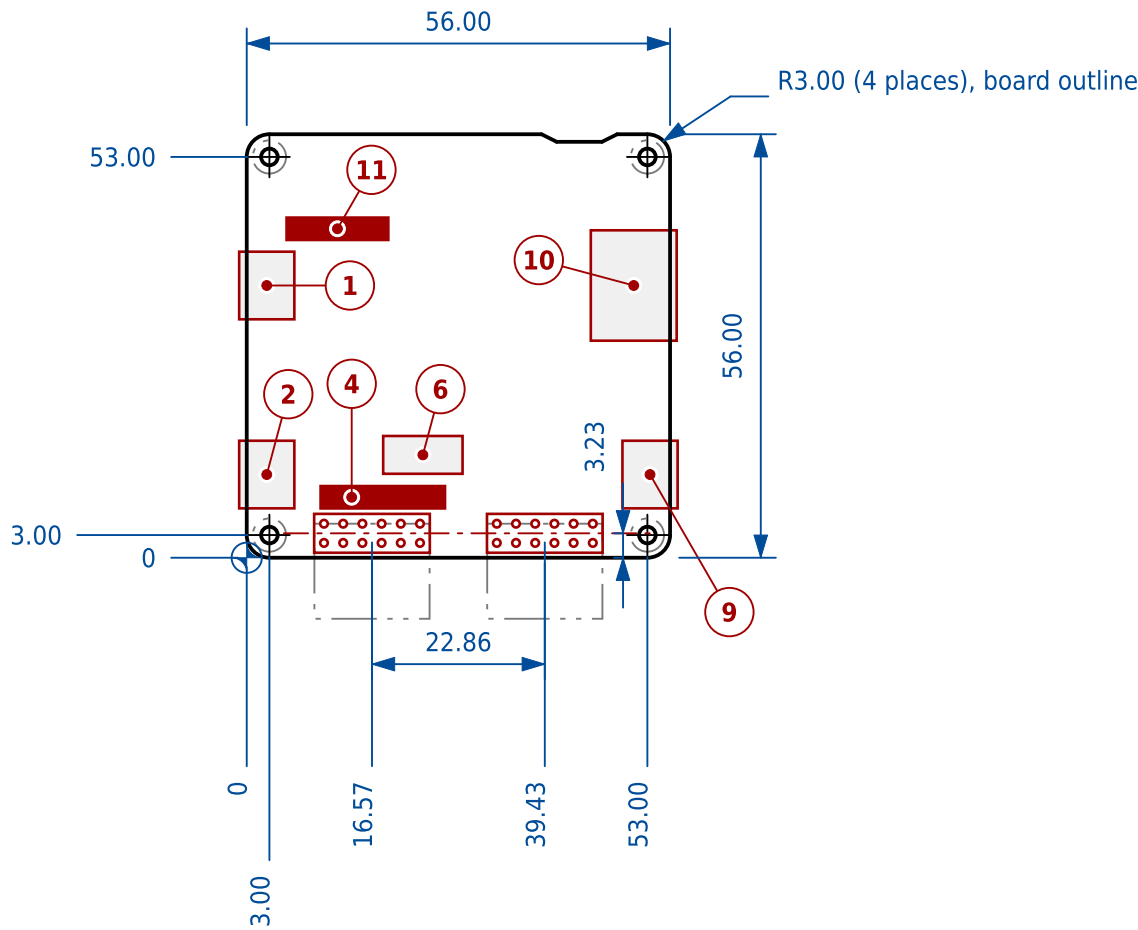
SOURCES

- KiCad board file: https://github.com/cheyao/icepi-zero-hardware/v1.3/icepi-zero.kicad_pcb @ 6e4aaba2 (tag v1.3) - title block: Icepi Zero rev v1.3, dated 2025-07-22; Chengyin Yao (cheyao). Solderpad Hardware Licence 2.1.
- Board revision: <https://github.com/cheyao/icepi-zero/releases/tag/v1.3> - the v1.3 tag, "Final mass production files": the revision made in quantity and sold. The same board file later in its history, at d67bb758, is read too and agrees on every position drawn here.
- Board size: <https://github.com/cheyao/icepi-zero/blob/v1.3/README.md> - quoted at this tag: "Icepi Zero is an FPGA development board in the popular Raspberry Pi Zero form factor"; the board file gives 65.00 x 30.00 mm.
- Assembly: <https://github.com/cheyao/icepi-zero/blob/v1.3/hardware/v1.3/production/bom.csv> - the production bill of materials at the same tag lists no 2x20 header: the GPIO position is not fitted.
- Form factor compared: <https://datasheets.raspberrypi.com/rpizero/raspberry-pi-zero-mechanical-drawing.pdf> - Raspberry Pi Zero, RPI-ZERO-V1_2 of 2015-09-23: 65 x 30 mm, corner radius 3.0, 4x M2.5 mounting holes drilled to 2.75 +/-0.05, 58 x 23 apart and 3.5 in from each edge.

ICEPI ZERO

v1.3, Raspberry Pi Zero form factor

DRAWN	VERSION	UNITS	SCALE
generated	v0.0-400-g80d77dd	mm	2:1
SIZE	SHEET	DRAWING NO	REV
A3	1 OF 1	FPGA-ICEPI-ZERO	A
MATERIAL	GENERAL TOLERANCE		
PCB, 1.6 nominal	edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08		



NOTES

1. Viewed from the component side.
2. Every dimension is read from the board file; Great Scott Gadgets publish no mechanical drawing. Hole IDs are the KiCad reference designators.
3. Schedule numbers 9, 10 and 11 were added to the family for this board. 1 to 8 mean what they mean on the other FPGA sheets, so the four USB ports read 1, 2, 9, 10.
4. No Ethernet jack.
5. Feature 6, J5, is a surface-mount mezzanine receptacle on the component side: a board-to-board expansion socket well inside the outline, not an edge port like the expansion connectors on the other sheets.
6. Pmod A and B are right-angle sockets: the pin field is on the board and the housing hangs off the front edge, so a peripheral plugs in level with the board rather than standing up from it. The housings reach 8.07 mm past the edge.
7. The board ships inside an enclosure. The outline drawn is the bare PCB's; the repository publishes no case dimensions.
8. Buttons PROGRAM and USER on the left edge and RESET on the right are not drawn. They are side-actuated tactile switches whose bodies reach 2.50 mm past the left edge and 2.50 mm past the right.
9. Assembled envelope, connector overhang and the Pmod housings included: 58.00 x 64.07 mm. With the three side buttons as well it is 61.00 x 64.07 mm, which is what a case has to clear.

NOTES (continued)

10. The top edge carries a recess 1.00 mm deep, flat from x = 41.00 to 47.00, each end ramping 2.00 back to the edge at x = 39.00 and 49.00. The board file gives no purpose for it.

SOURCES

1. KiCad board file:
https://github.com/greatscottgadgets/cynthion-hardware/cynthion.kicad_pcb @ 13aa71c2 (tag r1.4.0) - title block: Cynthion rev 1.4.0, Copyright 2019-2024 Great Scott Gadgets, resolved from the project file's text variables at the same commit; Great Scott Gadgets.
2. Revision sold:
<https://github.com/greatscottgadgets/cynthion-hardware/releases/tag/r1.4.0> - the r1.4.0 release notes read "Initial production release."; it is the newest release and the tip of the repository, and no commit since has touched the board file.
3. Licence:
<https://github.com/greatscottgadgets/cynthion-hardware/blob/r1.4.0/LICENSE> - CERN Open Hardware Licence Version 2 - Permissive, Copyright (c) 2019-2024 Great Scott Gadgets.
4. Ports and indicators:
https://cynthion.readthedocs.io/en/latest/hardware/device_overview.html - names the four USB ports CONTROL, AUX, TARGET C and TARGET A, in that order; six user LEDs driven by the FPGA and five status LEDs driven by the debug microcontroller.

HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
H1	3.00	3.00	2.20	4.40
H2	3.00	53.00	2.20	4.40
H3	53.00	53.00	2.20	4.40
H4	53.00	3.00	2.20	4.40

PMOD HOST HEADERS

PORT	REF	EDGE	CX mm	CY mm	PIN1 X mm	PIN1 Y mm
Pmod A	J7	bottom	16.570	3.230	22.920	4.500
Pmod B	J8	bottom	39.430	3.230	45.780	4.500

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	USB-C J2, CONTROL port	-1.00 to 6.30	31.53 to 40.47
2	USB-C J1, AUX port	-1.00 to 6.30	6.53 to 15.47
3	Ethernet RJ45 - not on this board	-	-
4	User LEDs D2-D7, 6 on 3.00 mm pitch	9.77 to 26.23	6.52 to 9.48
5	User LEDs, tri-colour - not on this board	-	-
6	Mezzanine receptacle J5, 30-way	18.05 to 28.55	11.10 to 16.10
7	Expansion connector, second - not on this board	-	-
8	Expansion connector, third - not on this board	-	-
9	USB-C J4, TARGET C port	49.70 to 57.00	6.53 to 15.47
10	USB-A J3, TARGET A port	45.52 to 56.88	28.70 to 43.30
11	Status LEDs D10-D14, 5 on 3.00 mm pitch	5.27 to 18.73	42.02 to 44.98

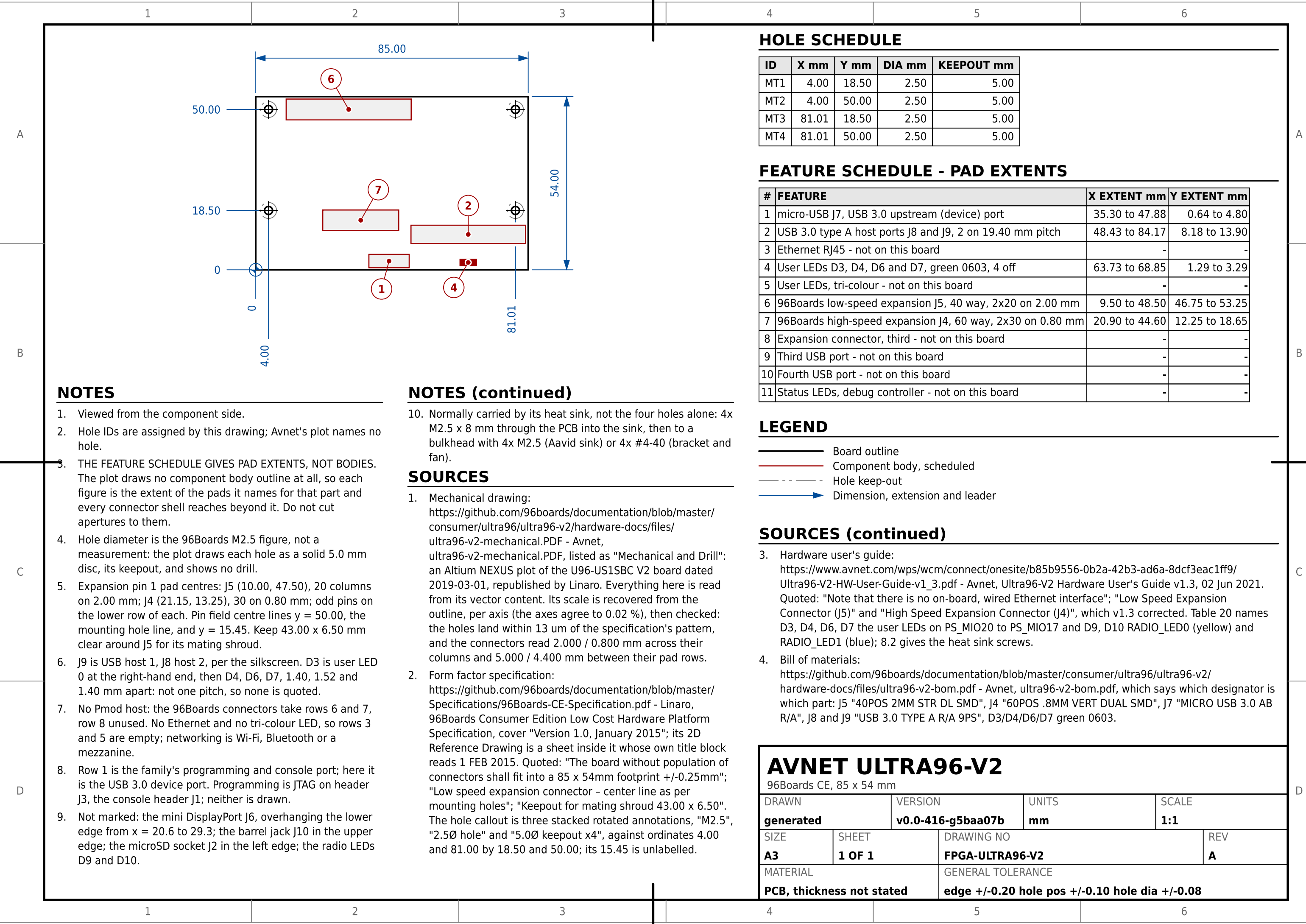
LEGEND

- Board outline
- Component body, scheduled
- Pmod connector body, hole keep-out
- Dimension, extension and leader

CYNTHION

r1.4.0

DRAWN	VERSION	UNITS	SCALE
generated	v0.0-430-gbfb1b8f	mm	1:1
SIZE	SHEET	DRAWING NO	REV
A3	1 OF 1	FPGA-CYNTHION	A
MATERIAL	GENERAL TOLERANCE		
PCB, 1.6 nominal	edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08		



NOTES

- 1. Viewed from the component side.
- 2. Hole IDs are assigned by this drawing; Avnet's plot names no hole.
- 3. THE FEATURE SCHEDULE GIVES PAD EXTENTS, NOT BODIES. The plot draws no component body outline at all, so each figure is the extent of the pads it names for that part and every connector shell reaches beyond it. Do not cut apertures to them.
- 4. Hole diameter is the 96Boards M2.5 figure, not a measurement: the plot draws each hole as a solid 5.0 mm disc, its keepout, and shows no drill.
- 5. Expansion pin 1 pad centres: J5 (10.00, 47.50), 20 columns on 2.00 mm; J4 (21.15, 13.25), 30 on 0.80 mm; odd pins on the lower row of each. Pin field centre lines y = 50.00, the mounting hole line, and y = 15.45. Keep 43.00 x 6.50 mm clear around J5 for its mating shroud.
- 6. J9 is USB host 1, J8 host 2, per the silkscreen. D3 is user LED 0 at the right-hand end, then D4, D6, D7, 1.40, 1.52 and 1.40 mm apart: not one pitch, so none is quoted.
- 7. No Pmod host: the 96Boards connectors take rows 6 and 7, row 8 unused. No Ethernet and no tri-colour LED, so rows 3 and 5 are empty; networking is Wi-Fi, Bluetooth or a mezzanine.
- 8. Row 1 is the family's programming and console port; here it is the USB 3.0 device port. Programming is JTAG on header J3, the console header J1; neither is drawn.
- 9. Not marked: the mini DisplayPort J6, overhanging the lower edge from x = 20.6 to 29.3; the barrel jack J10 in the upper edge; the microSD socket J2 in the left edge; the radio LEDs D9 and D10.

NOTES (continued)

- 10. Normally carried by its heat sink, not the four holes alone: 4x M2.5 x 8 mm through the PCB into the sink, then to a bulkhead with 4x M2.5 (Aavid sink) or 4x #4-40 (bracket and fan).

SOURCES

- 1. Mechanical drawing: <https://github.com/96boards/documentation/blob/master/consumer/ultra96/ultra96-v2/hardware-docs/files/ultra96-v2-mechanical.PDF> - Avnet, ultra96-v2-mechanical.PDF, listed as "Mechanical and Drill": an Altium NEXUS plot of the U96-US1SBC V2 board dated 2019-03-01, republished by Linaro. Everything here is read from its vector content. Its scale is recovered from the outline, per axis (the axes agree to 0.02 %), then checked: the holes land within 13 um of the specification's pattern, and the connectors read 2.000 / 0.800 mm across their columns and 5.000 / 4.400 mm between their pad rows.
- 2. Form factor specification: <https://github.com/96boards/documentation/blob/master/Specifications/96Boards-CE-Specification.pdf> - Linaro, 96Boards Consumer Edition Low Cost Hardware Platform Specification, cover "Version 1.0, January 2015"; its 2D Reference Drawing is a sheet inside it whose own title block reads 1 FEB 2015. Quoted: "The board without population of connectors shall fit into a 85 x 54mm footprint +/-0.25mm"; "Low speed expansion connector - center line as per mounting holes"; "Keepout for mating shroud 43.00 x 6.50". The hole callout is three stacked rotated annotations, "M2.5", "2.5Ø hole" and "5.0Ø keepout x4", against ordinates 4.00 and 81.00 by 18.50 and 50.00; its 15.45 is unlabelled.

HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
MT1	4.00	18.50	2.50	5.00
MT2	4.00	50.00	2.50	5.00
MT3	81.01	18.50	2.50	5.00
MT4	81.01	50.00	2.50	5.00

FEATURE SCHEDULE - PAD EXTENTS

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	micro-USB J7, USB 3.0 upstream (device) port	35.30 to 47.88	0.64 to 4.80
2	USB 3.0 type A host ports J8 and J9, 2 on 19.40 mm pitch	48.43 to 84.17	8.18 to 13.90
3	Ethernet RJ45 - not on this board	-	-
4	User LEDs D3, D4, D6 and D7, green 0603, 4 off	63.73 to 68.85	1.29 to 3.29
5	User LEDs, tri-colour - not on this board	-	-
6	96Boards low-speed expansion J5, 40 way, 2x20 on 2.00 mm	9.50 to 48.50	46.75 to 53.25
7	96Boards high-speed expansion J4, 60 way, 2x30 on 0.80 mm	20.90 to 44.60	12.25 to 18.65
8	Expansion connector, third - not on this board	-	-
9	Third USB port - not on this board	-	-
10	Fourth USB port - not on this board	-	-
11	Status LEDs, debug controller - not on this board	-	-

LEGEND

- Board outline
- Component body, scheduled
- Hole keep-out
- Dimension, extension and leader

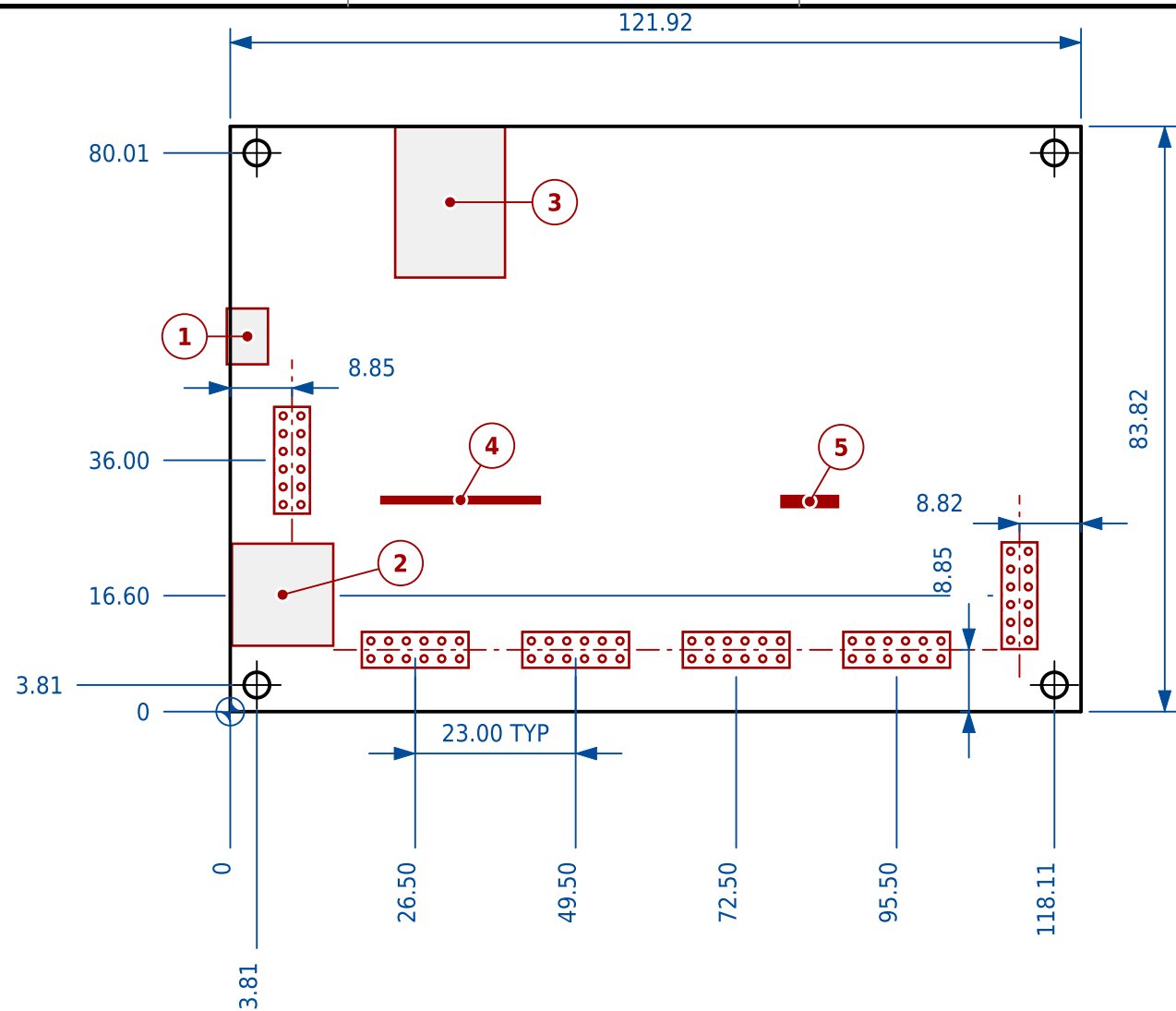
SOURCES (continued)

- 3. Hardware user's guide: https://www.avnet.com/wps/wcm/connect/onesite/b85b9556-0b2a-42b3-ad6a-8dcf3eac1ff9/Ultra96-V2-HW-User-Guide-v1_3.pdf - Avnet, Ultra96-V2 Hardware User's Guide v1.3, 02 Jun 2021. Quoted: "Note that there is no on-board, wired Ethernet interface"; "Low Speed Expansion Connector (J5)" and "High Speed Expansion Connector (J4)", which v1.3 corrected. Table 20 names D3, D4, D6, D7 the user LEDs on PS_MIO20 to PS_MIO17 and D9, D10 RADIO_LED0 (yellow) and RADIO_LED1 (blue); 8.2 gives the heat sink screws.
- 4. Bill of materials: <https://github.com/96boards/documentation/blob/master/consumer/ultra96/ultra96-v2/hardware-docs/files/ultra96-v2-bom.pdf> - Avnet, ultra96-v2-bom.pdf, which says which designator is which part: J5 "40POS 2MM STR DL SMD", J4 "60POS .8MM VERT DUAL SMD", J7 "MICRO USB 3.0 AB R/A", J8 and J9 "USB 3.0 TYPE A R/A 9PS", D3/D4/D6/D7 green 0603.

AVNET ULTRA96-V2

96Boards CE, 85 x 54 mm

DRAWN generated		VERSION v0.0-416-g5baa07b	UNITS mm	SCALE 1:1	
SIZE A3	SHEET 1 OF 1	DRAWING NO FPGA-ULTRA96-V2			REV A
MATERIAL PCB, thickness not stated		GENERAL TOLERANCE edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08			



NOTES

1. Viewed from the component side.
2. Hole IDs are this drawing's; the DXF names nothing.
3. Pin rows 2.50 mm apart on the drawing, not 2.54; the host pitch dimensioned below is not the specification's 22.86 (0.90 in).
4. Drawn fully fitted, as the Zybo Z7-20; the Z7-10 leaves Pmod JB and one tri-colour LED off the same board.
5. J10, a micro-AB USB socket UNDER J11, projects 2.72 mm below the board and 0.82 past the left edge; C250, under the Ethernet jack, goes deepest at 3.10.
6. HDMI, audio, Pcam, microSD, the power jack and external JTAG are not marked, so the envelope below counts only what is drawn; the model's, with the audio jacks 1.71 past the right edge, is 124.44 x 83.90.
7. Bodies read from the PDF plot are good to about +/-0.3 mm.

NOTES (continued)

8. Assembled envelope, connector overhang included: 122.42 x 83.82 mm.

SOURCES

1. Mechanical drawing: https://digilent.com/reference/_media/reference/programmable-logic/zybo-z7/zybo_z7_dimensions.zip - Digilent, "Zybo Z7 Mechanical Drawings", dated 2020-09-03. Edge, holes, pin fields, pegs and shell pads from the DXF; bodies from the PDF plot, its scale recovered from the outline per axis (the two agree to 0.20 %).
2. 3D model: https://files.digilent.com/resources/programmable-logic/zybo-z7/Zybo_Z7.step - Digilent, Zybo_Z7.step, in the drawing's frame: its solids carry the designators, so they name the plot's outlines. Laminate 1.571 mm.

SOURCES (continued)

3. Reference manual: https://digilent.com/reference/_media/reference/programmable-logic/zybo-z7/zybo-z7_rm.pdf - Revised 2018-02-21, section 16: JA is the XADC port, JB, JC and JD high-speed, JE standard, JF the MIO port; four user LEDs, and "the Zybo Z7-10 only has one tri-color LED".
4. Schematic: <https://files.digilent.com/resources/programmable-logic/zybo-z7/zybo-z7-d1-sch.pdf> - Revision D.1: J12 is the micro-USB PROG/UART port, J11 the "USB A" host and J10 a "USB Micro AB" beneath it.
5. Resource Center: <https://digilent.com/reference/programmable-logic/zybo-z7/start-Physical>, quoted: "Width 3.3 in (88 mm)" and "Length 4.8 in (122 mm)"; the drawing gives 121.92 x 83.82, exactly 4.8 x 3.3 in. Its top view of the board has 3V3 and GND silkscreened at each host's end farthest from pin 1.

HOLE SCHEDULE

ID	X mm	Y mm	DIA mm	KEEPOUT mm
MT1	3.81	3.81	3.66	not given
MT2	118.11	3.81	3.66	not given
MT3	3.81	80.01	3.66	not given
MT4	118.11	80.01	3.66	not given

PMOD HOST HEADERS

PORT	EDGE	CX mm	CY mm	PIN1 X mm	PIN1 Y mm
JA	right	113.100	16.600	111.850	22.950
JB	bottom	95.500	8.850	101.850	10.100
JC	bottom	72.500	8.850	78.850	10.100
JD	bottom	49.500	8.850	55.850	10.100
JE	bottom	26.500	8.850	32.850	10.100
JF	left	8.850	36.000	10.100	29.650

FEATURE SCHEDULE

#	FEATURE	X EXTENT mm	Y EXTENT mm
1	micro-USB J12, programming and console	-0.50 to 5.41	49.74 to 57.74
2	USB type A host port J11	0.24 to 14.76	9.44 to 24.05
3	Ethernet RJ45 J3	23.63 to 39.38	62.17 to 83.75
4	LEDs LD0-LD3, 4 on 7.00 mm pitch	21.65 to 44.37	29.84 to 30.76
5	Tri-colour LEDs LD5 and LD6, 2 on 6.50 mm pitch	78.99 to 87.10	29.29 to 30.89
6	Expansion connector, first - not on this board	-	-
7	Expansion connector, second - not on this board	-	-
8	Expansion connector, third - not on this board	-	-
9	Third USB port - not on this board	-	-
10	Fourth USB port - not on this board	-	-
11	Status LEDs, debug controller - not on this board	-	-

LEGEND

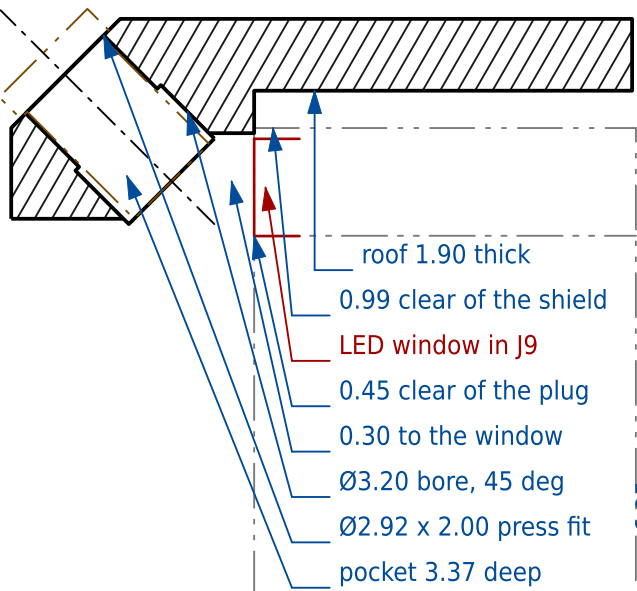
- Board outline
- Component body, scheduled
- Pmod connector body
- Dimension, extension and leader

DIGILENT ZYBO Z7

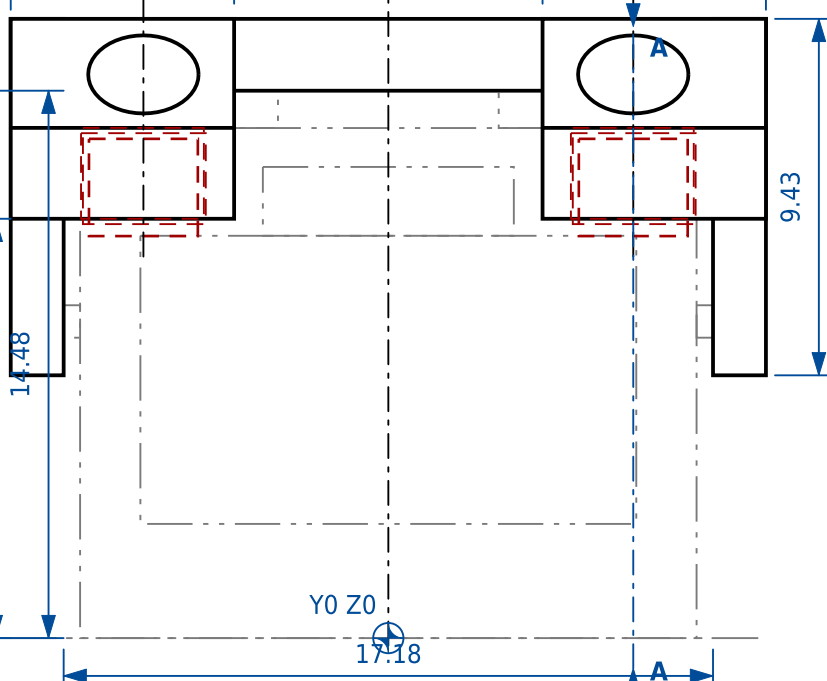
Z7-10 and Z7-20

DRAWN generated	VERSION v0.0-430-gbfb1b8f	UNITS mm	SCALE 1:1
SIZE A3	SHEET 1 OF 1	DRAWING NO FPGA-ZYBO-Z7	REV A
MATERIAL PCB, 1.6 nominal	GENERAL TOLERANCE edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08		

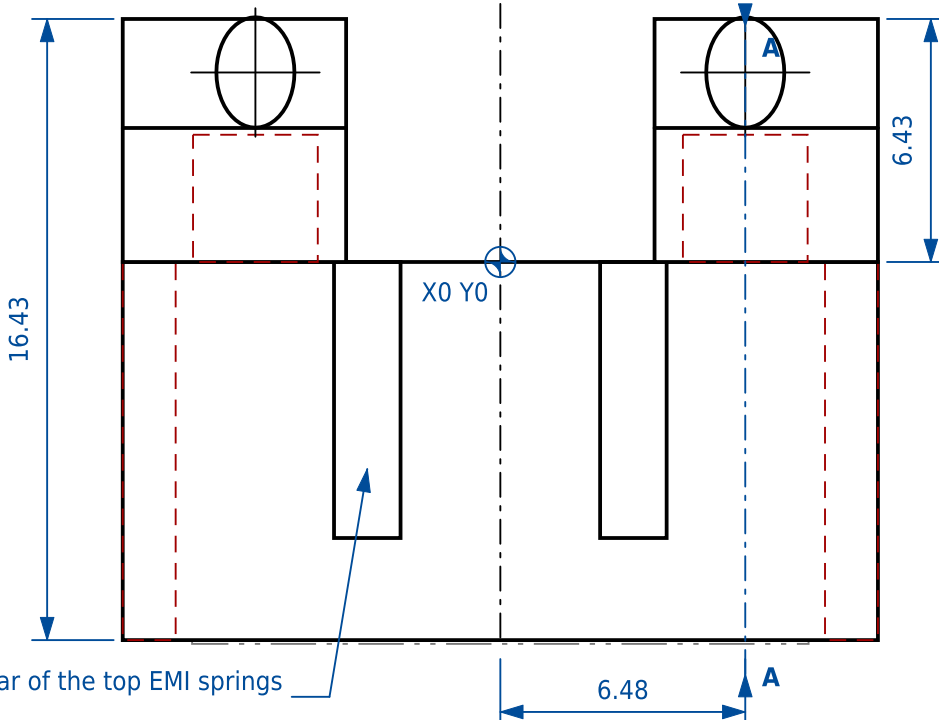
SECTION A-A



FRONT ELEVATION



PLAN



BORE SCHEDULE, DERIVED

ID	Y mm	TIP X / Z mm	DIA mm	PRESS mm	PRESS LEN	DEG
B1, B2	+/-6.48	-2.18 / 12.08	3.20	2.92	2.00	45

THE JACK, J9 08B0-1X1T-36-F

FEATURE	Y mm	Z mm	READ AS
LED window, 0.64 proud	5.04..7.92	10.63..13.20	MEASURED
Plug aperture	+/-6.56	3.02..10.64	MEASURED
Latch keyway, widest	+/-3.32	10.64..12.46	MEASURED
Shield	+/-8.15	0..13.49	STATED
EMI spring, side	+/-9.18	7.95..8.80	MEASURED
EMI spring, top	+/-2.92..4.12	to 14.52	MEASURED

LEGEND

- The printed part
- Hidden detail: bores and the pocket
- The RJ45 J9 and its EMI springs, adjacent parts
- The two Bivar light pipes, bought

NOTES (continued)

- The press fit is Bivar's $\varnothing 2.92 +0.08/-0.05$ and that governs, not the general tolerance; ream it if the printed hole comes out undersize.
- A plug and a plain boot pass under the cheeks; a snagless boot standing proud of the plug's own top face within 7 mm of the jack will foul them.
- STATED is Bel's own dimension, ± 0.254 ; MEASURED is read off its geometry, ± 0.20 ; DERIVED is worked out from those and Bivar's pipe.

SOURCES

- Board schematic: https://diligent.com/reference/_media/reference/programmable-logic/art7/art7_sch.pdf - Diligent, Arty A7 rev E.0, sheet 8 of 12 (ETHERNET): J9 is 08B0-1X1T-36-F.
- Jack drawing: <https://www.belfuse.com/media/drawings/products/magjack%20ICMs/dr-mag-08b0-1x1t-36-f.pdf> - Bel 08B01X1T36-F rev E; MEASURED figures are scaled off its front and side views.
- Board drawing: https://diligent.com/reference/_media/reference/programmable-logic/art7/art7.zip - Diligent, Arty_A7_DXF.DXF: board locks 16.104 apart against Bel's 16.13; centre plane $y = 44.0$, and through Bel's 7.75 the front face $x = -0.101 \pm 0.628$, which is how far Diligent's own plot of the jack disagrees.
- Light pipe drawing: https://www.bivar.com/parts_content/Datasheets/PLP2-XXX.pdf - Bivar PLP2-XXX rev Y: $\varnothing 2.8$ pipe on $\varnothing 3.1$ ribs, $\varnothing 3.3$ flange, $\varnothing 0.115$ in hole, 0.047-0.093 in panel.

NOTES

- First angle; SECTION A-A is on the right-hand bore's axis. Origin on the jack's centre plane, in its front face, at the board's top surface: X back, Y across, Z up. Symmetric about $Y = 0$.
- Two Bivar PLP2-4MM pipes in clear polycarbonate, pressed in from the facet until the flange seats. Each lens looks up and forward at 45 degrees, so a camera looking straight down sees it obliquely.

NOTES (continued)

- J9's LED windows stand 0.64 mm proud of the face the cheeks land on, so the pipe tips are set from the window, at maximum material.
- Nothing holds the adapter on but J9's own side EMI springs, 1.4 ± 0.51 proud of a 16.31 ± 0.254 shield, bearing on the skirts: deflected 0.33 mm at minimum material and 1.6 at maximum, the shield never touching.
- Print upside down, on the top face: the facets and the bores are then 45 degree overhangs and nothing needs support.

ARTY A7 ETHERNET LIGHT PIPE

Clips over J9 and carries its two LEDs to a face you can see from above

DRAWN generated		VERSION v0.0-468-g7b8e30e	UNITS mm	SCALE 5:1
SIZE A3	SHEET 1 OF 1	DRAWING NO FPGA-LP-ARTY		REV A
MATERIAL 3D printed, opaque		GENERAL TOLERANCE printed ± 0.20 bores: see note on the press fit		PROJECTION